

Demonstration of Proposed Features

Date	02 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Demonstration of Proposed Features :

S.N o	Feature Name	Description	Status	Demonstrat ed	Remarks
1	Data Loading & Integration	Successfully loaded and merged application and credit record datasets using the customer ID.	Implement ed	Yes	Dataset imported and verified successfully.
2	Data Preprocessin g	Performed missing value handling, feature engineering, categorical encoding, and feature scaling.	Implement ed	Yes	Clean dataset prepared for machine learning.
3	Machine Learning Model Training	Trained multiple classification models including Logistic Regression, Decision Tree, Random Forest, KNN, and SVM.	Implement ed	Yes	Random Forest selected as the best-performi ng model.

4	Model Evaluation	Evaluated models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC metrics.	Implemented	Yes	Performance comparison completed successfully.
5	Model Optimization	Performed Hyperparameter Tuning and Feature Importance Analysis to improve model performance.	Implemented	Yes	Model optimized for better predictive accuracy.
6	Model Saving & Prediction	Saved the trained model using Joblib for future credit approval predictions.	Implemented	Yes	Model is reusable for future deployment and inference.

Feature Implementation Summary :

Metric	Value
Total Features Proposed	6
Total Features Implemented	6
Total Features Demonstrated	6
Overall Implementation Rate (%)	100%